Team 67- LLOYDS B: Final Report

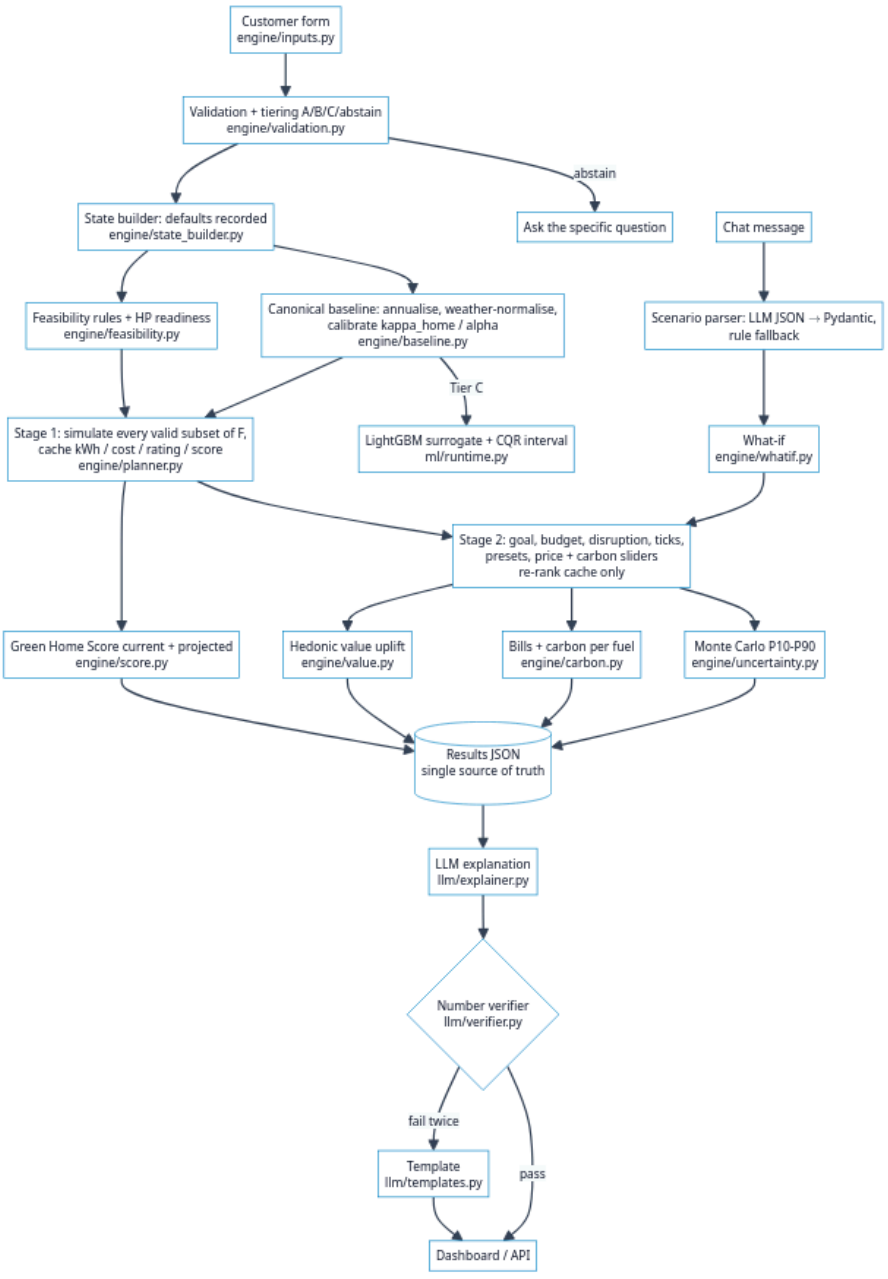
1. Solution

Our AI system answers four questions for a homeowner: *How green is my home today? What should I improve? How much energy, money and carbon could I save? Could it raise my property's value?* The customer fills in a short form. Every question accepts "not sure", and energy readings are optional. The system then returns:

- A Green Home Score out of 100, its class, and the reasons behind it.
- A prioritised list of improvements that actually suit the home.
- Savings per fuel, bill savings in ₹, carbon reduction and payback, each with a likely range.
- An indicative property-value range.
- A chat assistant for "what if" questions.

Core principle: every number comes from documented models and calculations. The AI language model (LLM) only explains results. A what-if question is first rerun through the engine, and a checker blocks any number the LLM did not receive from the engine.

2. Approach and Architecture



Data: No public dataset links home features, metered energy, retrofits and savings, so we built a physics-based simulator.

- 30,000 realistic homes generated from region, type, age, size, building features and heating.
- Monthly energy use includes heat loss, weather, heating, hot water, appliances, solar and batteries.
- Added realistic uncertainty: 4.2% missing months and ~8% “not sure” answers.
- Each retrofit and full package is simulated to capture interactions, such as insulation improving heat-pump efficiency.
- Calibrated to national consumption statistics; label rules are in `docs/data_schema.md`.

Technical Approach

- **Data splits:** Homes are split by customer with no leakage: **17,097 train, 3,658 validation, 3,679 test**, plus **4,462 OOD homes** from an unseen region and 1900–1929 building era. Only customer-provided information is used as input.
- **Engine:** Each home is validated and assigned a confidence tier. Missing or conflicting key data triggers a specific question. Usage is annualised and weather adjusted, then calibrated to meter readings. The engine scores the home, filters infeasible retrofits, simulates up to **512 combinations**, and caches them for fast re-ranking. Savings, carbon and property-value ranges are reported with uncertainty.
- **Machine learning:** LightGBM predicts energy use from incomplete inputs, correcting the physics estimate. Conformal prediction provides **80% ranges**. Additional models handle feasibility, savings previews, retrofit ranking and improvement potential.
- **LLM:** Gemini converts questions into validated scenarios, the engine reruns them, and the LLM explains the resulting facts. A **number verifier** blocks unsupported numbers, with template fallback after a failed rewrite. The provider can be swapped and backup models/providers are supported.
- **Stack:** Python, pandas, LightGBM, scikit-learn, FastAPI, Streamlit and Plotly. **60 automated tests** pass, with reproducibility through `make all` and fixed seeds.

3. Green Home Score (Sub-Score Mappings): $S = \Sigma\, w_i \cdot s_i$, using the required weights. The score is rounded to the nearest integer. Every mapping gives a better state an equal or higher sub-score, and this is checked by unit tests.

- **Energy efficiency (25%):** Input: Recorded EPC rating or SAP points; estimated if unknown. ***Sub-score mapping*** (0-100): Band A 100, B 85, C 70, D 55, E 40, F 25, G 10 (interpolated on SAP points).
- **Energy intensity (20%)**: Input: Annualised, weather-adjusted metered kWh per m². ***Sub-score mapping*** (0-100): ≤50→100, 100→80, 150→60, 250→30, 350→10, ≥450→0.
- **Insulation (15%):** Input: Loft, walls, floor. ***Sub-score mapping*** (0-100): 0.4·roof + 0.4·wall + 0.2·floor (e.g. loft <100 mm 35, >250 mm 100; cavity unfilled 35, filled 80).
- **Heating (15%):** Input: System, boiler age. ***Sub-score mapping*** (0-100): Heat pump 100; gas boiler <10 yrs 65, 10–15 yrs 55, >15 yrs 45; electric 30.
- **Glazing (10%)**: Input: Window type. ***Sub-score mapping*** (0-100): Triple 100, modern double 80, old double 55, single 20.
- **Renewables (10%):** Input: Solar generation ÷ electricity demand. ***Sub-score mapping*** (0-100): 0% → 0, rising to 100 at 50%.
- **Smart controls (5%):** Input: Controls. ***Sub-score mapping*** (0-100): None 0, timer 30, smart thermostat 60, smart radiator valves 80.

Classes: 80–100 Highly efficient · 65–79 Good · 50–64 Moderate · 30–49 Poor · 0–29 Very poor.

Avoiding double counting. Efficiency uses the *recorded rating*, while intensity uses *actual metered use per m²*. Solar is rewarded only in the renewables component, because intensity uses total consumption including solar used on site. The dashboard shows each component's points earned out of points available, so any overlap is visible.

4. Features

- **Explainable score:** Gauge, component-wise points earned vs available, and top issues costing the most points.

- **Home ranking:** Poor and Very poor homes are flagged and ranked by improvement potential, based on percentage reduction in grid electricity and gas.
- **Improvement plan:** Suitable improvements are ordered logically: **insulation** → **heating** → **solar** → **controls**, with suitability, reasons, prerequisites and fuel-specific savings.
- **Plan tailoring:** Users can set goals, budget in ₹, disruption tolerance, and include/exclude choices. A cost-versus-saving chart compares plans.
- **Heat-pump honesty:** The app clearly explains increased electricity use alongside reduced gas use.
- **What-if:** Energy price and grid-carbon sliders update the best plan, with chat support for questions like adding a heat pump or reaching Band C.
- **Customer memory:** Completed or declined improvements affect future recommendations.
- **Other:** Light/dark mode, ₹ currency, and REST API endpoints for score, plan, what-if, explain and assumptions.

5. Results: Results are measured against our simulator, not real-world accuracy.

Validation results:
- MAE: Electricity 21.0% vs baseline 54.1%; Gas 16.5% vs 63.6%
- R^2: Electricity 0.88; Gas 0.79
- 80% range coverage: 0.80 for both
- Test / OOD error: Electricity 20.9% / 21.4%; Gas 16.4% / 16.6%

Other held-out results:
- Package savings MAE: 160 kWh electricity, 408 kWh gas
- Improvement ranking: Spearman 0.97
- Retrofit ranking: NDCG@3 0.98
- Feasibility: 98% precision / 98% recall
- Planner: Optimal, 7 ms plan and 21 ms re-rank
- Property value error: 14.9%
- Electricity error across regions/types/ages stayed within 19.5–23.8%

Key findings and predictions: The ML closely matches the physics model, while physics inputs improve performance. Results are sensitive to simulator settings, and value ranges cover 73% of cases. The LLM number checker passed 100%, but groundedness (70.2%), accuracy (78.9%) and unsupported claims (5.1%) missed targets. `test_predictions.csv` contains predictions and reference labels; `portfolio_ranking.csv` ranks homes by improvement potential. 31% were Poor/Very poor and none required more information. Definitions are in `docs/data_schema.md`.

6. Responsible AI
- **Privacy:** Uses only synthetic data; no names or addresses are sent to the LLM, and API keys stay out of the repository.
- **Bias:** Checks errors across region, property type and age band, including unseen regions and eras. Ranking uses percentage improvement rather than home size.
- **Explainability:** Fixed score mappings, reasons for recommendations and exclusions, and per-fuel savings.
- **Hallucination control:** Validated scenarios, engine reruns, number verification and template fallback. LLM evaluation results are reported openly.
- **Human oversight:** Estimates include confidence tiers and ranges. The app asks for clarification rather than making assumptions, recommends qualified assessors, and notes that tenants need the landlord's consent.
- **Data limitations:** Labels are simulated; prices, costs and emission factors are placeholders; the rating is an EPC-style estimate rather than an official EPC; value uplift is observational rather than causal; and ₹ amounts use a placeholder exchange rate rather than local market prices.